Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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Abstract

Resumo

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Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is food waste, a global stream that in the European Union alone is estimated at 88 million tonnes per year, excluding inedible parts removed from the supply chain (Scherhaufer et al., 2018). When considering only post-processing activities – wholesale, retail, food service and households – approximately 66.3 million tonnes of food waste are produced annually in the EU-27+1 (Albizzati et al., 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of food waste is substantial. It accounts for roughly 15–16% of the total climate impact of the entire food supply chain, translating to about 186 Mt CO_2-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhaufer et al., 2018). The majority of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favoured over landfill disposal (Albizzati et al., 2021; Scherhaufer et al., 2018). Despite this, much food waste continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

Parallel to the food waste challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: crude glycerol. Formed in the transesterification of triglycerides, approximately 10 kg of crude glycerol are produced for every 100 kg of biodiesel (Garlapati et al., 2016). Global biodiesel output, driven by renewable energy mandates, has caused the glycerol market to swell – with crude glycerol supply projected to reach about 6.33 million tonnes by 2025, of which nearly 680 000 tonnes are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachi et al., 2023).

Crude glycerol composition varies widely depending on the oil source and catalyst used, but it typically contains methanol, soaps, free fatty acids, salts and other matter organic non-glycerol (MONG) compounds (Anand & Saxena, 2012; Attarbachi et al., 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-propanediol production when untreated crude glycerol from jatropha, soybean or sunflower oils was used (Anand & Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachi et al., 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, crude glycerol has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher et al., 2025; Tomczak et al., 2025).

Anaerobic digestion (AD) is a mature biological process in which consortia of microorganisms convert organic matter into biogas – primarily methane (55–65%) and carbon dioxide – under oxygen-free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis and methanogenesis (Gebreegziabher et al., 2025). AD offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertiliser; and reduces the environmental load of organic wastes by avoiding landfilling and its associated methane emissions (Gebreegziabher et al., 2025; Scherhaufer et al., 2018).

Within this framework, anaerobic co-digestion – the simultaneous treatment of two or more substrates – has emerged as a particularly effective strategy. Co-digesting crude glycerol with nitrogen-rich materials such as animal manure, sewage sludge or food waste balances the carbon-to-nitrogen ratio, dilutes potential inhibitors, provides essential macro- and micronutrients, and fosters synergistic microbial interactions (Gebreegziabher et al., 2025; Tomczak et al., 2025). The result can be a dramatic improvement in biogas and methane yield: studies report increases ranging from 14% to 226% when crude glycerol is added to a variety of base substrates, with methane contents generally exceeding 60% as long as glycerol does not exceed about 8% (v/v) of the feed (Gebreegziabher et al., 2025; Tomczak et al., 2025).

However, excessive glycerol loads can lead to volatile fatty acid accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher et al., 2025; Tomczak et al., 2025). Overall, the convergence of abundant food waste and surplus crude glycerol supplies, together with the proven capability of anaerobic digestion to convert these streams into renewable methane and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this dissertation project is to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic food waste and glycerol as substrates. The investigation will be conducted by determining biogas production, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of glycerol, a by-product of the biodiesel industry, enhances biogas production due to its high energy content and rapid biodegradability, provided that adequate operational conditions are maintained to avoid volatile fatty acid overload.

2.2 Specific Objectives

1. Characterize the substrates (food waste and glycerol), evaluating physicochemical parameters relevant to anaerobic digestion.

2. Operate the single-stage anaerobic co-digestion system, assessing process stability.

3. Evaluate the techno-economic feasibility of using glycerol as a co-substrate, considering energy yield and potential for industrial application.

Chapter 3

Literature Review

The rapid expansion of biodiesel production has generated an oversupply of crude glycerol, a by-product of the transesterification output, this image 3.1 shows the steps to produce biodiesel and when glycerol is producesd (Garlapati et al., 2016). This waste stream, loaded with impurities such as methanol, soaps, and hydroxides, the image 3.2 shows the processes and what they contaminate the glycerol with, poses environmental disposal challenges, also with the variation of impurities depending on the source, there’s no standard method to use this source of carbon (Anand & Saxena, 2012; Gebreegziabher et al., 2025). Anaerobic digestion (AD) aligns with circular economy principles by converting organic residues into biogas while recycling nutrients, making crude glycerol a promising co-substrate (Gebreegziabher et al., 2025).

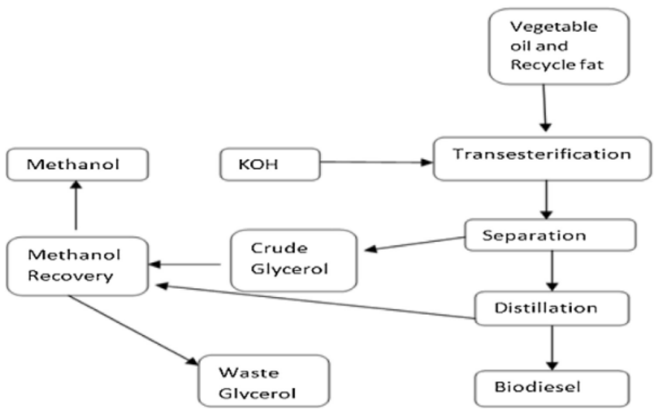

Figure 3.1: Source: Garlapati et al. (2016)

Gebreegziabher et al. (2025) reported that co-digestion of crude glycerol can boost methane production compared to mono-digestion of the base substrate. Tomczak et al. (2025) reviewed one-stage AD experiments and concluded that a methane content above 60% could be consistently obtained when the crude glycerol fraction did not exceed 8% (v/v) of the feed, the flow chart by Li et al. (2011) on the image 3.3 it shows that using

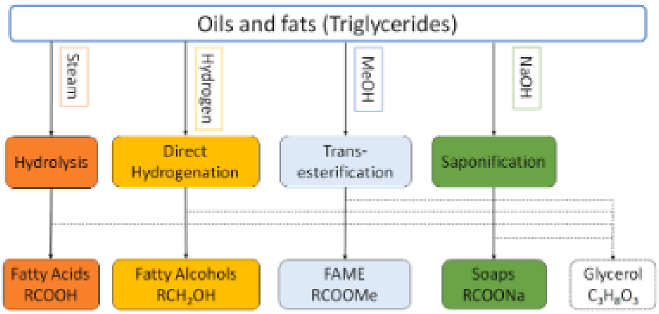

Figure 3.2: Source: Attarbachi et al. (2023)

AD there could be up to 70% methane production. However, the same review emphasized that increasing the glycerol loading beyond a system-specific threshold may provoke inhibitory effects, likely due to the accumulation of volatile fatty acids (VFAs) and the impurities present in crude glycerol (Anand & Saxena, 2012; Tomczak et al., 2025). The heterogeneous composition of crude glycerol, affected by oil feedstock, catalyst, and biodiesel processing, demands careful feedstock characterization and, if necessary, simple pretreatment to avoid process upsets (Attarbachi et al., 2023).

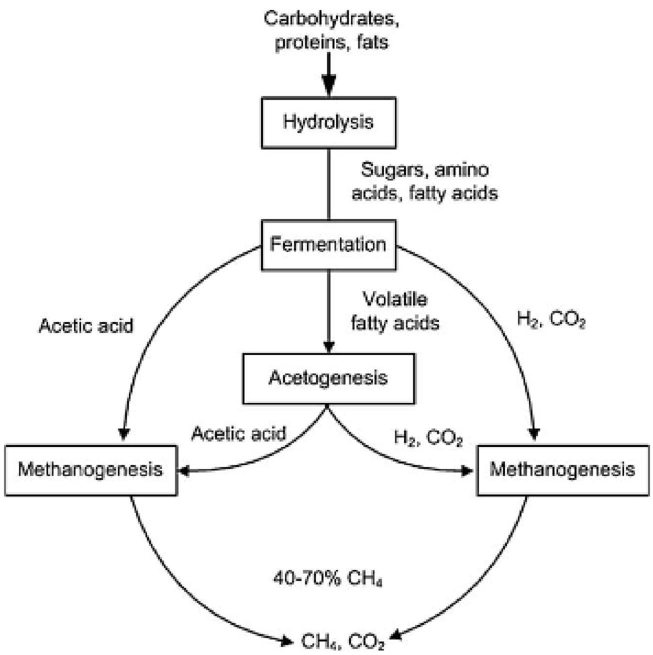

Figure 3.3: Source: Li et al. (2011)

Liquid anaerobic digestion (AD), typically conducted in a continuous stirred tank reactor (CSTR), operates at total solid concentrations of 0.5–15% and is generally used for treating sewage sludge, manure, and food waste (Li et al., 2011). The CSTR's continuous mixing provides uniform distribution of substrates, microorganisms, and heat, which is essential when feeding a fermentable substrate. Temperature is a critical operational parameter; the activity of methanogenic consortia is strongly temperature dependent, and both mesophilic and thermophilic regimes have been applied to AD (Choorit & Wisarnwan, 2007). Tomczak et al. (2025) noted that the majority of glycerol co-digestion studies have been performed under mesophilic conditions, while thermophilic operation, which could offer higher conversion rates, still requires more systematic investigation. The CSTR platform is well suited to explore such temperature effects because its design allows easy control of heating and hydraulic retention time.

The selection of a robust microbial inoculum is decisive for rapid start-up and stable performance of a CSTR fed with glycerol. Anaerobic sludge from municipal wastewater treatment plants (WWTPs) is the most common source of inoculum because it harbours a diverse consortium of hydrolytic, acidogenic, and methanogenic microorganisms, the flow cahrd from Gebreegziabher et al. (2025) on the image 3.4 shows each step of the AD until the methane production. Posmanik et al. (2020) demonstrated that granular sludge, obtained from up-flow anaerobic sludge blanket reactors treating wastewater, outperforms suspended biomass as an inoculum. Although their work focused on batch tests, the findings strongly suggest that using granular sludge from a WWTP as seed for a CSTR could accelerate acclimation, improve process stability, and reduce the risk of acidification during load increases.

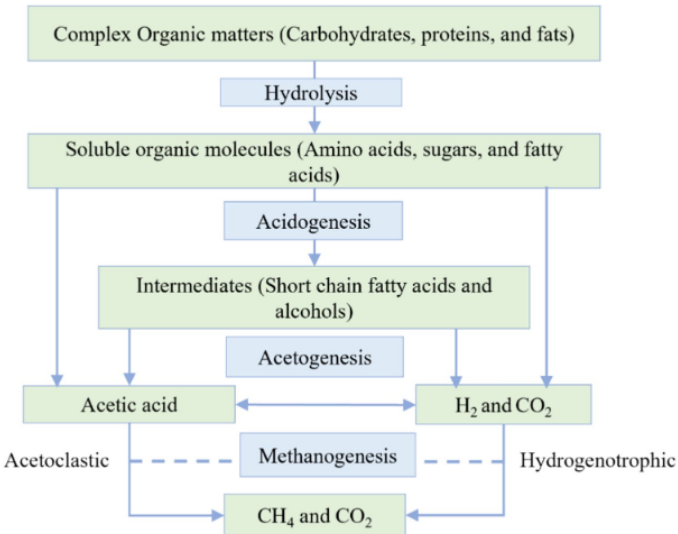

Figure 3.4: Source: Gebreegziabher et al. (2025)

When glycerol starts being degraded it makes the AD process susceptible to VFA accumulation and pH drop if the organic loading rate is not properly controlled. Therefore, close monitoring of process indicators is essential. Ribas et al. (2007) evaluated several methods for determining these parameters specifically in anaerobic reactors, highlighting the importance of accurate VFA and alkalinity measurements for timely detection of process imbalance. Using the method from DiLallo & Albertson (Ribas et al., 2007) provide a validated method for measuring VFAs and alkalinity. Maintaining a favourable VFA-to-alkalinity ratio is particularly critical when co-digesting glycerol, as excess VFAs can inhibit methanogenesis (Tomczak et al., 2025).

The available literature confirms that crude glycerol is a high-energy substrate that can significantly boost methane production in liquid AD systems. A CSTR provides the mixing and temperature control required to harness this potential. Using granular sludge from a wastewater treatment plant as the inoculum can enhance start-up speed and methanogenic activity, as demonstrated by comparative studies. To unlock the full benefits while avoiding inhibition, the glycerol loading must be kept below documented thresholds, and the reactor state should be continuously tracked through standard measurements of VFAs and alkalinity. Further research is needed to optimise thermophilic CSTR operation and to establish long-term performance data for granular-sludge-inoculated reactors treating crude glycerol.

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

Food waste was collected from the canteen at the Instituto Politécnico de Bragança (IPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize the sugar content and minimize the nitrogen content. The waste was diluted with approximately 2.5 L of distilled water and homogenised using a blender to obtain a liquid paste. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce et al., 2020).

4.2 Crude Glycerol Collection

A 6 L bottle of crude glycerol, obtained from previous biodiesel production experiments at IPB, was provided for this study. Crude glycerol from biodiesel typically contains impurities such as catalyst residues (Attarbachi et al., 2023), which may influence its behaviour in anaerobic digestion.

4.3 Inoculum Collection

The anaerobic inoculum consisted of sludge collected from Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for the anaerobic digestion (Posmanik et al., 2020).

4.4 Physical and Chemical Characterisation

The total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the food waste, crude glycerol, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for the food waste and crude glycerol.

4.4.1 TS and VS Determination

TS and VS were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Baird et al., 2017). The method involves controlled drying and calcination steps to quantify total solids and volatile solids fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. The crucibles were then dried in an oven at 105 °C for 2 h (crude glycerol required 6–8 h to reach constant mass). After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The same crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Afterwards they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

TS, VS and FS were calculated as follows:

$$\text{TS} = \frac{M_{105} - M_{\text{crucible}} * 1000}{V_{\text{sample}}} \tag{4.1}$$

$$\text{VS} = \frac{M_{105} - M_{550} * 1000}{V_{\text{sample}}} \tag{4.2}$$

$$\text{FS} = \frac{M_{550} - M_{\text{crucible}} * 1000}{V_{\text{sample}}} \tag{4.3}$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, M_{crucible} is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in mL and the weight is in mg. The results are on the table 4.1.

Table 4.1: TS, VS and FS results

Sample		M_{crucible}	M_{wet}	M_{sample}	V_{sample}	M_{105}	TS (g/L)	Average TS (g/L)	M_{550}	VS (g/L)	Average VS (g/L)	FS (g/L)	Average FS (g/L)
Food Waste	1	78.7310	105.9875	27.2565	25.5	91.0021	481.0194		79.2024	462.5408		18.4786	
	2	84.3206	113.2979	28.9773	0.0271	98.8684	536.3998	536.3998	84.7435	520.8068	520.8068	15.5930	18.4786
	3	77.5887	108.8399	31.2512	0.0292	93.4417	541.9933		78.1848	521.6134		20.3799	
Glycerol	1	84.2193	88.6434	4.4241	0.0050	88.4966	863.4402		84.2144	864.4293		-0.9891	
	2	73.4780	77.9874	4.5094	0.0050	77.8594	867.7240	867.7240	73.4733	868.6548	868.6548	-0.9308	-0.9891
	3	86.2012	90.6107	4.4095	0.0049	90.5034	871.3421		86.1958	872.4358		-1.0937	
Inoculum	1	78.7310	105.9875	27.2565	0.0255	91.0021	481.0194		79.2024	462.5408		18.4786	
	2	84.3206	113.2979	28.9773	0.0271	98.8684	536.3998	536.3998	84.7435	520.8068	520.8068	15.5930	18.4786
	3	77.5887	108.8399	31.2512	0.0292	93.4417	541.9933		78.1848	521.6134		20.3799	

Reference: Author, 2026

Table 4.2: Density results

Sample		M_{cylinder}	M_{total}	V_{sample}	Density (g/L)	Average Density (g/L)
Food Waste	1	136.0000	242.0000	0.0992	1068.7905	1068.4376
	2	116.0000	223.0000	0.1002	1068.0847	
Glycerol	1	114.2100	203.4700	0.0999	893.2353	893.0740
	2	115.7400	204.4400	0.0993	892.9127	
Inoculum	1	235.0000	242.0000	0.0992	1068.7905	1068.4376
	2	216.0000	223.0000	0.1002	1068.0847	

Reference: Author, 2026

4.4.2 Density Determination

Density was determined by a gravimetric–volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed again to verify the reading accuracy (density of water at 20 °C = 998.21 g L^{-1}). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \tag{4.4}$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume, the masses were messured in g and the volume in L. The results are on the table 4.2.

4.4.3 TOC and TN Determination

Total organic carbon (TOC), total carbon (TC) and total nitrogen (TN) were measured using a Shimadzu TOC-5000A analyser. Prior to analysis, the food waste was diluted 400-fold and the crude glycerol was diluted 800-fold to bring the concentrations within the linear range of the instrument. The results are on the table 4.3.

4.5 Experiment Configuration

The reactor had a useful volume of 1 L and was operated as a continuous stirred-tank reactor (CSTR) under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow exit tube was kept closed with a valve to maintain an anaerobic environment.

Table 4.3: TS, VS and FS results

Sample		TOC (mg/L)	TC (mg/L)	TN (mg/L)	Dilution	TC (mg/L)	Average TC (mg/L)	TN (mg/L)	Average TN (mg/L)
Food Waste	1	5.03900	5.04400	0.68910	400	2017.6	1778.6	275.64	281.76
	2	3.92700	3.84900	0.71970		1539.6		287.88	
Glycerol	1	8.80800	8.81200	0.58750	800	7049.6	5523.2	470	488.6
	2	4.99200	4.99600	0.63400		3996.8		507.2	

Reference: Author, 2026

The reactor was kept at 35 °C, close to the optimal temperature for mesophilic microorganisms (Choorit & Wisarnwan, 2007). The organic loading rate (OLR) was set with a max of 4 kg VS m^{-3} d^{-1}, with a hydraulic retention time (HRT) of 30 days, conditions for stable co-digestion of food waste and glycerol (Tomczak et al., 2025). The pH was maintained around 7 by incorporating $KHCO_3$ into the feed solution at a concentration of 10 g L^{-1}, and adding $KHCO_3$ to the reactor whenever the pH dropped lower than 6.

The organic loading rate (OLR) and hydraulic retention time (HRT):

$$\text{HRT} = \frac{V}{Q} \quad [\text{d}] \tag{4.5}$$

$$\text{OLR} = \frac{S_{\text{in}} \times Q}{V} \quad [\text{kg VS m}^{-3}\,\text{d}^{-1}] \tag{4.6}$$

where S_{in} is the influent VS concentration (kg m^{-3}), Q is the volumetric flow rate (m^3 d^{-1}), and V is the reactor working volume (m^3).

4.5.1 Feeding Solution

The feed was designed to achieve a carbon-to-nitrogen (C/N) ratio of 25:1, which is optimal for anaerobic digestion (Li et al., 2011). Characterisation of the food waste and crude glycerol showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both of them together. Therefore, pure glycerol was used as an additive to allow precise control of the C/N ratio. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid food-waste suspension through the feeding tubes.

The feed was made with a restriction of needing to have less than 15% of TS%. The amount of OLR and if $KHCO_3$ was added, was decided by the pH and how was the methane production of the reactor. If it was too acidic, it was used a lower OLR with $KHCO_3$, if it was at the right amount it was keep the same amount, and if it was too basic it was added a higher OLR without $KHCO_3$. The feed storage always had some remainig feed, and the new one was added on top, diluting or concentrating the feed going to the reactor.

4.5.2 Biogas Quantification

It was used the *bioprocess control AMPTS II (Automatic Methane Potential Test System)* system, to meassure the gas flow that was comming out of the reactor, with a CO_2 fixing unit using a solution of 3 M NaOH.

4.5.3 Analysis Methods

It was used the method of DiLallo & Albertson (Ribas et al., 2007), to measure the alkalinity and volatile fatty acids. The sample collected was first centrifuged, at 2604 G for 30 minutes, to filter the particulated before doing the measurements.

$$\mathrm{TA} = \frac{N_{\mathrm{acid}} * V_{\mathrm{acid}} * 50000}{V_{\mathrm{sample}}} \tag{4.7}$$

$$\mathrm{VFA} = \frac{N_{\mathrm{base}} * V_{\mathrm{base}} * 60000}{V_{\mathrm{sample}}} \tag{4.8}$$

The volumes were measured in mL, the sample volume was always 50 mL, the normality (N) of the acid and base, at first was 0.1, but then it was changed to 1. The equation 4.7 is expressed in $mgCaCO_3/L$, and the equation 4.8 is expressed in $mgCH_3COOH/L$.

Chapter 5

Results

Chapter 6

Discussion

Chapter 7

Conclusion

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